



## Namal University Mianwali

### Department of Computer Science

#### 1. COURSE DETAILS

|                         |                             |
|-------------------------|-----------------------------|
| Course Title:           | Web Application Development |
| Course Code:            | CS-370                      |
| Credits (Theory + Lab): | 3 (3+0)                     |
| Lecture Hours/Week:     | 3 hours                     |
| Pre-Requisite:          | -                           |

**Brief Course Description:** This course provides a comprehensive introduction to web application development, focusing on both frontend and backend technologies. Students will learn the fundamentals of HTML, CSS, JavaScript, and React for building responsive and dynamic user interfaces. On the backend, the course covers Node.js, Express.js, and API creation, as well as database integration using MongoDB. By the end of the course, students will be equipped with the skills to develop, optimize, and secure full-stack web applications.

#### 2. INSTRUCTOR DETAILS

|                  |                                      |
|------------------|--------------------------------------|
| Name:            | Mr. Ammar Ahmad Khan                 |
| Lecture Timings: | Wed (Last 2 Slots)                   |
| Office Hours:    | Mon, Tue, Wed (3 <sup>rd</sup> Slot) |
| Office Location: | Office # 16, CS Faculty Offices      |
| Office Phone:    | +92 459 236995 Ext. 173              |
| Email:           | ammar.ahmad@namal.edu.pk             |

#### 3. COURSE UNIT WISE MAJOR TOPICS

| Unit No.            | Topics                                                | No of Teaching Hours |
|---------------------|-------------------------------------------------------|----------------------|
| 1                   | Fundamentals of Web Development                       | 1.5                  |
| 2                   | HTML Essentials                                       | 3                    |
| 3                   | Styling with CSS and Bootstrap                        | 6                    |
| 4                   | JavaScript Basics and Dynamic Web Development         | 6                    |
| 5                   | Frontend Frameworks with React                        | 7.5                  |
| 6                   | Backend Development with Node.js and Express.js       | 9                    |
| 7                   | REST API Development and Frontend-Backend Integration | 4.5                  |
| 8                   | MongoDB and Full Stack Deployment                     | 7.5                  |
| Total Contact Hours |                                                       | 45                   |

#### 4. COURSE LEARNING OUTCOMES (CLOs)

| Sr. No | Unit No   | CLO Statement                                                                | Taxonomy Level | GA |
|--------|-----------|------------------------------------------------------------------------------|----------------|----|
| CLO-1  | 1,2,3,4,5 | <b>Implement</b> web development <u>concepts</u> to build website            | C3             | 2  |
| CLO-2  | 2- 8      | <b>Design and develop</b> web applications that meet specified requirements. | C5             | 4  |
| CLO-3  | 2-8       | <b>Use</b> modern web development tools and technologies.                    | C3             | 5  |

## 5. WEEKLY LECTURE BREAKUP

| Lecture #            | Unit # | Topics Covered in Class                                                                                                                              | Reference in Book / Assessments |
|----------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| 1                    | 1      | Course overview, Web Basic, Client-Server Model, Web Communication Protocol (HTTP), Web Generations, and Standards & Constraints.                    | TB1                             |
| 2                    | 2      | Introduction to HTML structure, Key HTML tags (headings, paragraphs, links, images, div), Creating a basic webpage with HTML, Semantic HTML concepts | TB1                             |
| 3                    | 2      | Forms in HTML (input types, buttons), Tables and lists in HTML, Multimedia elements (audio, video), Accessibility features in HTML                   | TB1                             |
| 4                    | 3      | CSS Fundamentals (Selectors, Properties, Box Model, and Styling)                                                                                     | TB1                             |
| 5                    | 3      | Advanced CSS (Flexbox, Grid Layout, Media Queries)                                                                                                   | TB1<br><b>Assignment 1</b>      |
| 6                    | 3      | Introduction to the Bootstrap framework, Setting Up Bootstrap Bootstrap grid system for responsive design, Utilities and Basic Styling               | TB1                             |
| 7                    | 3      | Using Bootstrap components (navbars, buttons, cards), Forms and Inputs, Customizing Bootstrap, Media queries for responsiveness                      | TB1                             |
| 8                    | 4      | Introduction to JavaScript: Variable Declaration, Data Types, Operators, and Adding JavaScript to HTML.                                              | TB2                             |
| 9                    | 4      | JavaScript: Conditional Statements, Loops, Functions, DOM Manipulation                                                                               | TB2                             |
| 10                   | 4      | Event handling (click, submit), ES6+ Features, Events, and Error Handling, JavaScript integration with HTML and CSS                                  | TB2                             |
| 11                   | 4      | Working with APIs (AJAX, Fetch API, and JSON)                                                                                                        | TB2<br><b>Assignment 2</b>      |
| 12                   | 5      | Introduction to Frontend Frameworks (Overview of React, Angular, Vue)                                                                                | TB2<br><b>Quiz 1</b>            |
| 13                   | 5      | React Basics (Components, Props, State).                                                                                                             | TB2                             |
| 14                   | 5      | Advanced React (Hooks, Context API).                                                                                                                 | TB2                             |
| 15                   | 5      | React Routing and State Management.                                                                                                                  | TB2                             |
| 16                   | 5      | Integrating React with Bootstrap, Frontend Development through react.js                                                                              | TB2                             |
| <b>Mid Term Exam</b> |        |                                                                                                                                                      |                                 |
| 17                   | 6      | Backend Development Overview, Introduction to Server-Side Development, Node.js, Express.js                                                           | TB2                             |

|                        |   |                                                                                                               |                                    |
|------------------------|---|---------------------------------------------------------------------------------------------------------------|------------------------------------|
| 18                     | 6 | Node.JS: Architectures & Core Features, and Modules.                                                          | TB2                                |
| 19                     | 6 | Node Architecture: Event loop, Callback                                                                       | TB2                                |
| 20                     | 6 | Express Framework Concepts: Middleware, Routing, and Routers                                                  | TB2<br><b>Quiz 2</b>               |
| 21                     | 6 | Application Architecture in Express.JS: Adding Routes, Router & Middleware.                                   | TB2                                |
| 22                     | 6 | Application Development using View Generators, and Adding Bootstrap                                           | TB2<br><b>Assignment 3</b>         |
| 23                     | 7 | REST Architecture Principals: Statelessness, Idempotent, Method Selection, Resource, and URI Identifications. | TB2                                |
| 24                     | 7 | Building a Simple REST API                                                                                    | TB2                                |
| 25                     | 7 | Frontend Integration with Backend API's                                                                       | TB2                                |
| 26                     | 8 | Introduction to Mongo DB, Creation, Collections, Schemas, and its Access with Aggregate Functions             | TB2<br><b>Quiz 3</b>               |
| 27                     | 8 | Mongo Database Design, Implementation of MongoDB Client Connector with Node.JS (Mongo Client)                 | TB2                                |
| 28                     | 8 | Mongoose Connector Schemas, Models & its CRUD Operations                                                      | TB2<br><b>Assignment 4</b>         |
| 29                     | 8 | Introduction to Deployment (Hosting Platforms, CI/CD)                                                         | TB2                                |
| 30                     | 8 | Deploying Full Stack Applications (Frontend and Backend).                                                     | TB2<br><b>Project Presentation</b> |
| 31                     | - | Project Exhibition                                                                                            |                                    |
| 32                     | - | Course Review                                                                                                 |                                    |
| <b>Final Term Exam</b> |   |                                                                                                               |                                    |

## 6. CLO ASSESSMENT MECHANISM

| Assessment Tool | CLO1 | CLO2 | CLO3 |
|-----------------|------|------|------|
| Quizzes (3)     | ✓    |      |      |
| Assignment (4)  |      | ✓    | ✓    |
| Project         |      | ✓    | ✓    |
| Mid Term Exam   | ✓    |      | ✓    |
| Final Term Exam | ✓    | ✓    |      |

## 7. TEACHING MATERIAL

### Text Books:

1. HTML & CSS QuickStart Guide by David DuRocher (2021)
2. "Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node" by Vasan Subramanian, 2<sup>nd</sup> Edition (2019)

### Reference Material:

1. Learning React: Functional Web Development with React And Redux by Alex Banks & Eve Porcello
2. w3schools.com

## 8. POLICY & PROCEDURES

- **Attendance Policy:** Students must attend 75% of this course's lectures. Students who fall short of the required attendance percentage cannot appear in the terminal examination.
- **Plagiarism Policy:** Namal University's rules and regulations prohibit plagiarism, copying, and other dishonest behavior. Violators will face serious disciplinary action.
- **Course Assessment:**

| Assessment Tool | Quiz (3) | Assignment | Project (CCP) | Mid Term Exam | Final Term Exam | Total |
|-----------------|----------|------------|---------------|---------------|-----------------|-------|
| Theory (T)      | 10       | 15         | 10            | 25            | 40              | 100   |

## 9. VERIFICATION

**(i) I verify that the content of this document is correct and up-to-date.**

|                                                                                                                                                 |                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| <br><u>Ammar Ahmad Khan</u><br>Instructor's Name and Signature | <u>17<sup>th</sup> Feb, 2026</u><br>Date |
|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|

**(ii) I have reviewed course-outline and state that it complies with Namal University policies and guidelines.**

|                                                                            |                                          |
|----------------------------------------------------------------------------|------------------------------------------|
| <u>Dr. Muhammad Ali Shahid</u><br>Name and Signature of Head of Department | <u>17<sup>th</sup> Feb, 2026</u><br>Date |
|----------------------------------------------------------------------------|------------------------------------------|